Competing bandits in Matching Markets

presented by Etienne Boursier

January 28, 2021
Matching Reading group

Outline

- Introduction to bandits problem
- Competing bandits in Matching Markets[1]
- Subsequent works[2][3]

[1] Liu, L. T., Mania, H., & Jordan, M. *AISTATS 2020.*

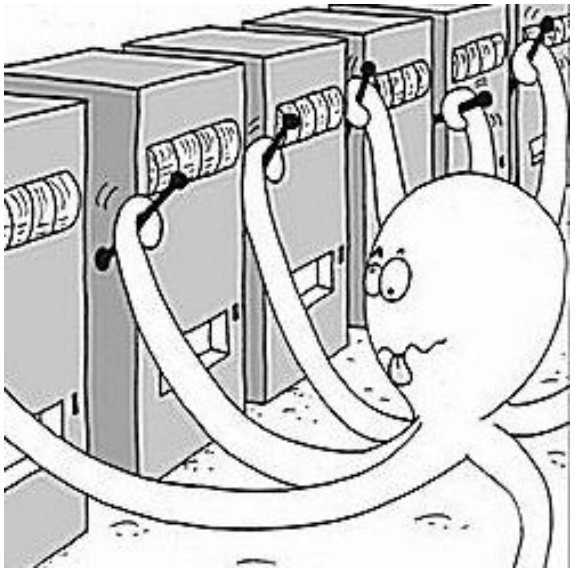
[2] Liu, L. T., Ruan, F., Mania, H., & Jordan, M. I. Bandit Learning in Decentralized Matching Markets

[3] Sankararaman, A., Basu, S., & Sankararaman, K. A. Dominate or Delete: Decentralized Competing Bandits with Uniform Valuation

Introduction to bandits problem

Multi-armed bandits (MAB)

- online learning problem
- widely used in online advertisement
- allows nice theoretical bounds
- many existing variations



Stochastic MAB

For $t = 1, \ldots, T$:

- choose arm $\pi(t)$ in $[K]$ to pull
- observe reward $X_{\pi(t)}(t) \in [0, 1]$ with X_k of mean μ_k (i.i.d.)

Notation: statistic order of means $\mu_{(1)} \geq \mu_{(2)} \geq \ldots \geq \mu_{(K)}$
Goal: maximize total reward or, equivalently, minimize regret

$$R(T) = \mu_{(1)} T - \sum_{t=1}^{T} \mathbb{E} X_{\pi(t)}$$

Exploration/exploitation dilemma: only observe reward of pulled arm

Explore-Then-Commit algorithm (ETC)

Lemma (Hoeffding inequality)

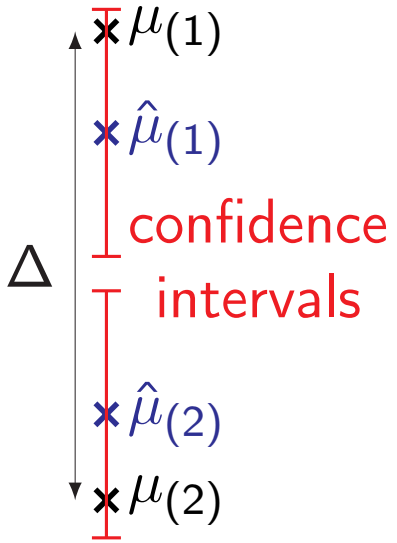
For X_i i.i.d. in $[0,1]$ of mean μ, $\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq \varepsilon\right) \leq 2e^{-2n\varepsilon^2}$.

Assume $\Delta = \mu_{(1)} - \mu_{(2)}$ known

Pull all arms $8\frac{\log(T)}{\Delta^2}$ times Pull best empirical arm until the end

With probability $1 - \frac{2K}{T}$
best empirical arm = true best arm
$\implies \quad R(T) \lesssim \frac{K\log(T)}{\Delta^2}$

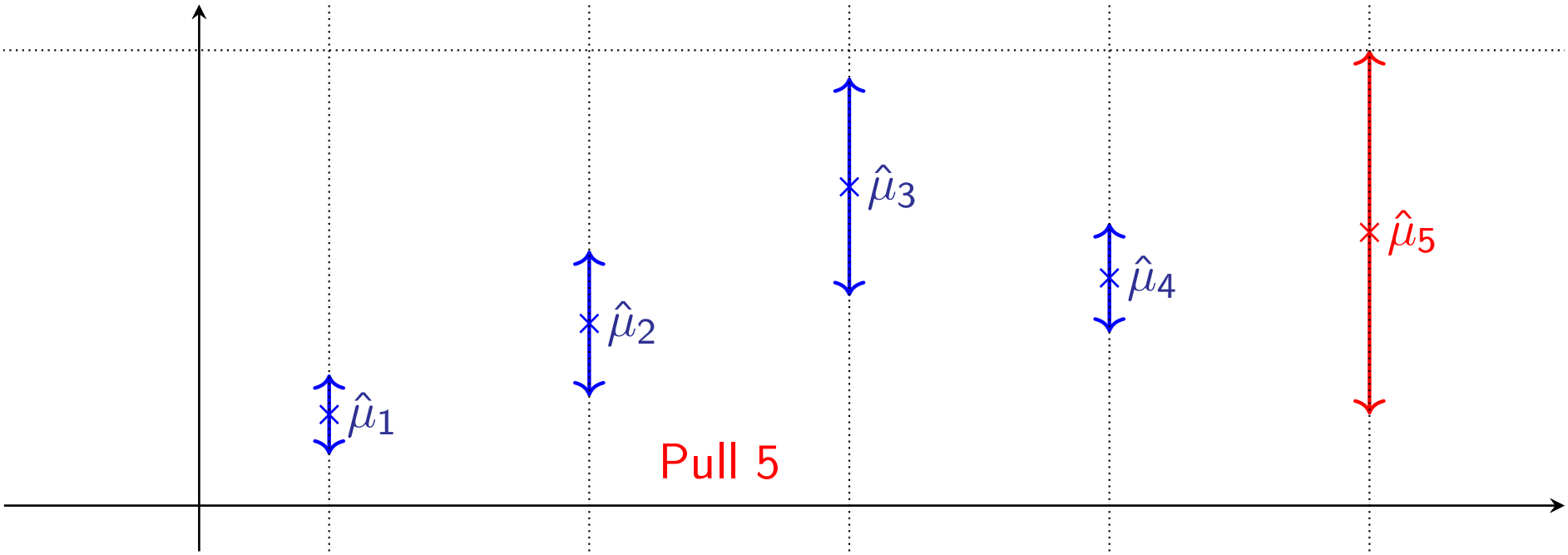
Drawback: require knowledge of Δ



Upper confidence bound algorithm (UCB)

Optimistic in face of uncertainty

Pull $\arg\max_{k\in[K]} \hat{\mu}_k + 2\sqrt{\frac{\log(T)}{N_k}}$



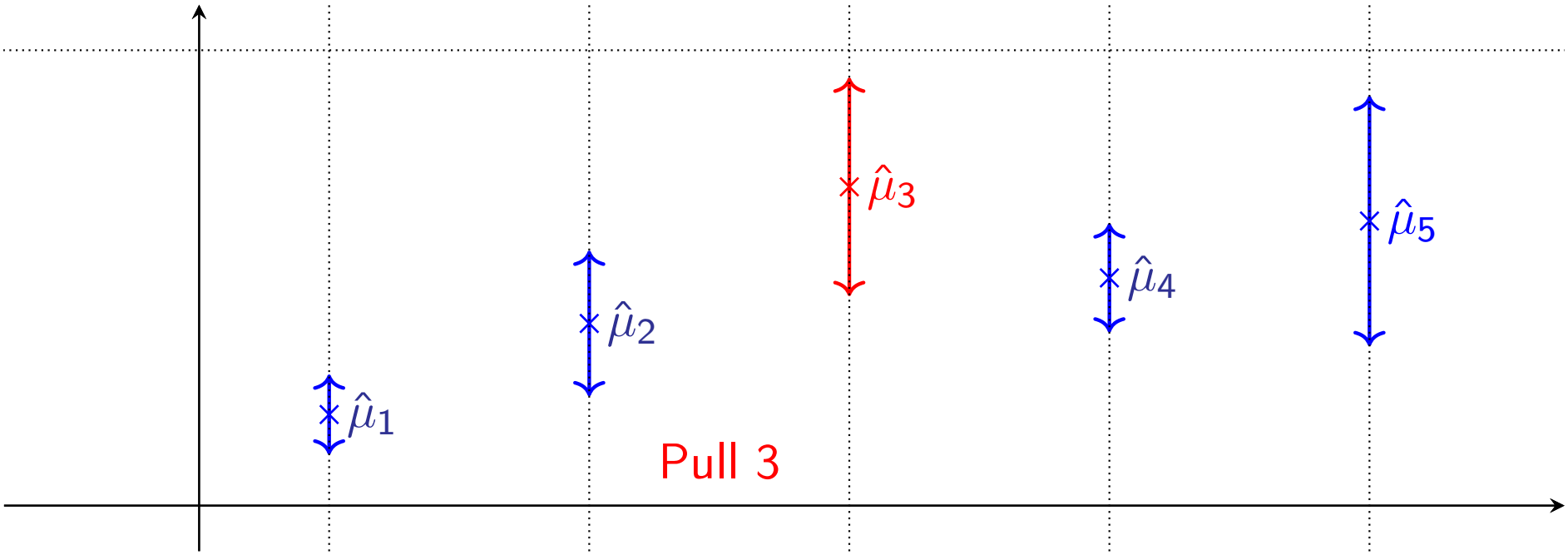
$$R(T) \lesssim \sum_{k>2} \frac{\log(T)}{\mu_{(1)} - \mu_{(k)}}$$

Optimal regret bound: no algorithm can do better

Upper confidence bound algorithm (UCB)

Optimistic in face of uncertainty

Pull $\arg\max_{k\in[K]} \hat{\mu}_k + 2\sqrt{\frac{\log(T)}{N_k}}$



$$R(T) \lesssim \sum_{k>2} \frac{\log(T)}{\mu_{(1)} - \mu_{(k)}}$$

Optimal regret bound: no algorithm can do better

Competing bandits in Matching Markets

by Lydia T. Liu, Horia Mania & Michael I. Jordan

Motivation

- Model at interface between ML and econ
- Markets as decentralized algorithms
- Two sided markets where consumers ignore their preferences
  $\rightarrow$ exploration/exploitation dilemma
- Competition between agents in markets
  $\rightarrow$ interaction between agents of same side

Model

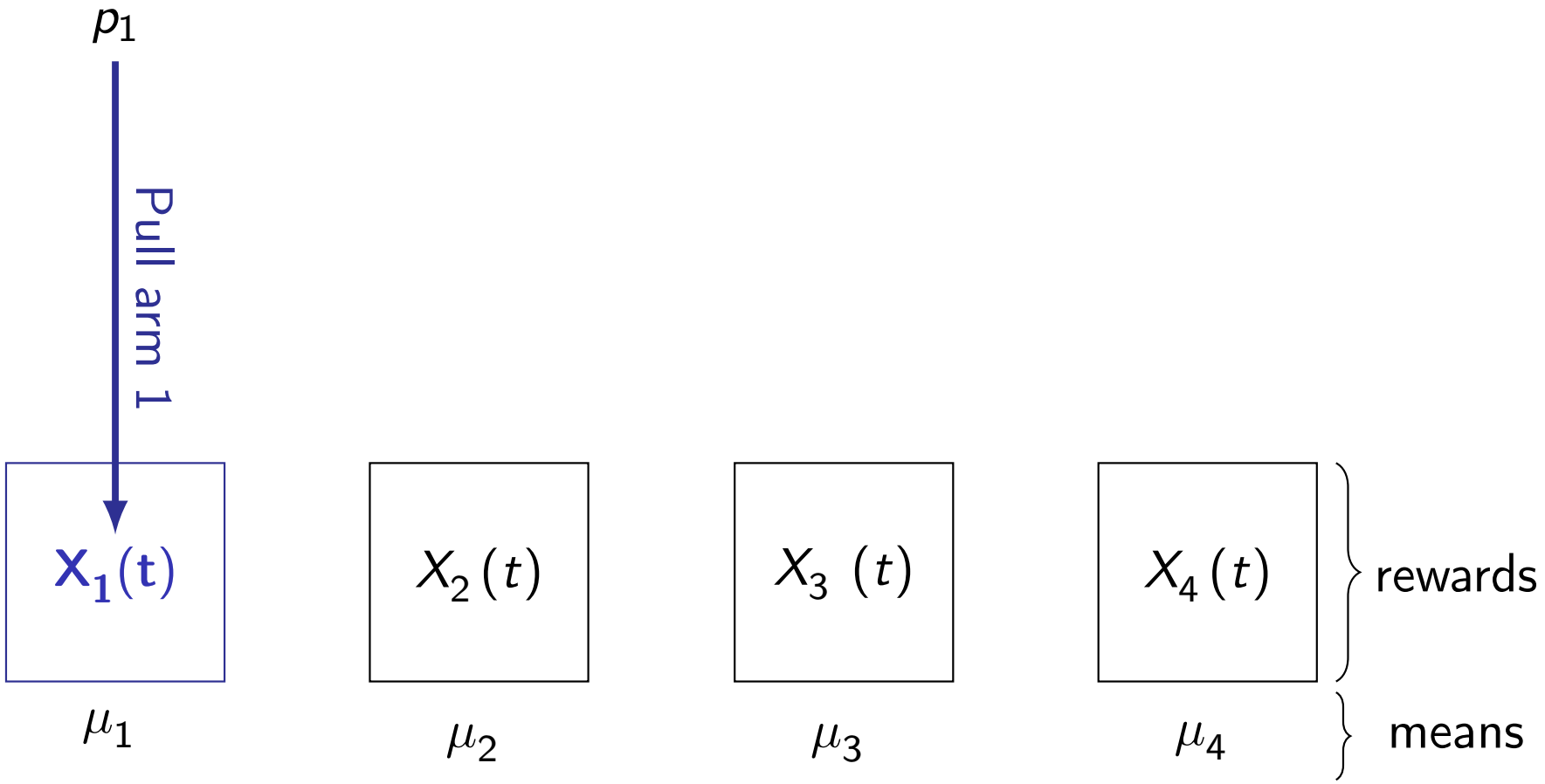
N players, K arms
arm $a_k \in \mathcal{K}$ has rewards $X_k^j(t) \in [0,1]$ of mean μ_k^j for player $p_j \in \mathcal{N}$
Arm ranking over players: $p_i \succ_k p_j \iff$ arm a_k prefers p_i over p_j

Each time $t = 1, \ldots, T$

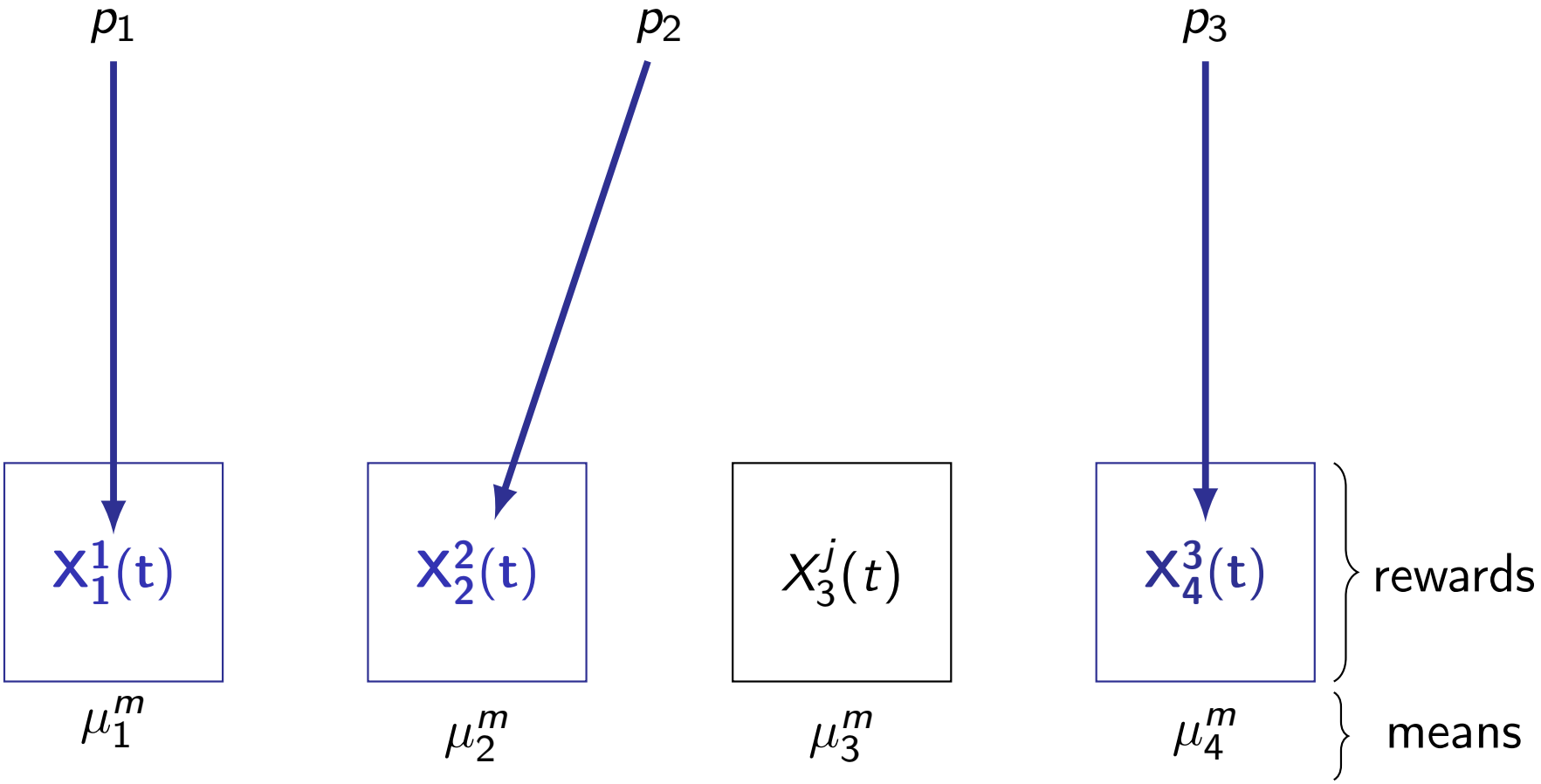
- each player p_i pulls arm $\pi_i(t) \in \mathcal{K}$
- receive and observe reward $r^i(t)$ defined as

$$r^i(t) = \begin{cases} X^i_{\pi_i(t)}(t) & \text{if } p_i \text{ is the most preferred (for } \pi_i(t)) \\ & \quad \text{player pulling } \pi_i(t) \\ 0 & \text{otherwise.} \end{cases}$$

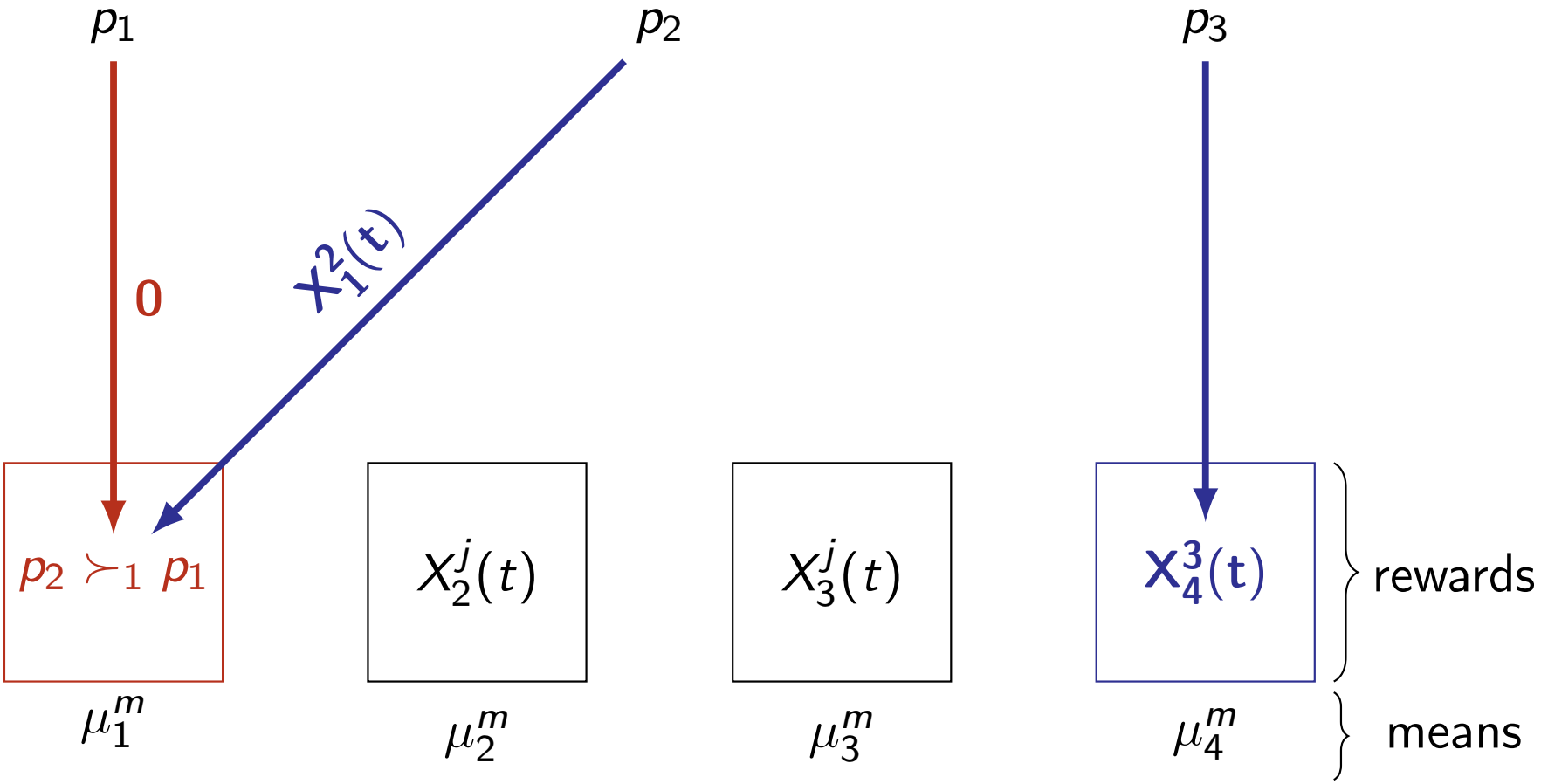
Stochastic bandits



Stochastic bandits [**Competing**]



Stochastic bandits [**Competing**]



Objectives

Stable matching π: no unmatched pair (p_i, a_k) prefers to be matched

Optimal matching: $\bar{\pi}_i$ best possible arm for player p_i in stable matching
Pessimal matching: $\underline{\pi}_i$ worst possible arm for player p_i in stable matching

Optimal regret for p_i: $\bar{R}_i(T) = \mu_{\bar{\pi}_i} T - \sum_{t=1}^{T} \mathbb{E} r^i(t)$

Pessimal regret for p_i : $\underline{R}_i(T) = \mu_{\underline{\pi}_i} T - \sum_{t=1}^{T} \mathbb{E} r^i(t)$

Decentralized Gale-Shapley algorithm

Input: preferences $\succ_k$ of arms over players
Input: preferences of players over arms
Initialize all arms and players to free; $n^i \leftarrow 1$ for all players
while *there is a free player* **do**
  all free players propose their n^i-th preferred arm
  arm a_k chooses the player it prefers among its current match and all proposing players in this round
  all free players increase n^i by 1
end
return *stable matching π between arms and players*

Gale-Shapley (GS) ends with optimal matching if players are on the proposing side!

First algorithm: Centralized ETC

Let $\bar{\Delta}_j^i = \mu_{\bar{\pi}_i}^i - \mu_j^i$ and $\bar{\Delta} = \min_i \min_{j:\bar{\Delta}_j^i > 0} \bar{\Delta}_j^i$

Pull all arms $\approx \frac{\log(T)}{\bar{\Delta}^2}$ times	`// exploration`
Run GS with players' rankings $\hat{\mu}_k^i$	
Pull arms assigned by GS until T	`// exploitation`

$\bar{R}_i(T) \lesssim \frac{K \log(T)}{\bar{\Delta}^2}$

Sketch of proof

1. After exploration, $\hat{\mu}_k^i \leq \hat{\mu}_{\bar{\pi}_i}^i$ for all k such that $\mu_k^i \leq \mu_{\bar{\pi}_i}^i$ with probability $1 - \frac{KN}{T}$
2. With such a ranking, $\bar{\pi}$ is a stable matching
   $\implies$ p_i gets assigned by GS an arm of mean $\geq \mu_{\bar{\pi}_i}^i$

First algorithm: Centralized ETC

Drawbacks

- Centralized for GS
  $\rightarrow$ decentralized GS takes small time if observe which player gets to pull the arm at each round
- Dependence in $\frac{1}{\Delta^2}$
- require knowledge of Δ

First algorithm: Centralized ETC

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Useful definitions

- For a matching π, a triplet $(p_j, a_k, a_{k'})$ is blocking when p_j is matched with $a_{k'}$, but both p_j and a_k prefer to be matched together
- Given a set S of matchings, a set Q of triplets is a cover of S if

$$S \subset \bigcup_{(p_j, a_k, a_{k'}) \in Q} B_{j,k,k'}.$$

  where $B_{j,k,k'}$ is the set of matchings blocked by the triplet $(p_j, a_k, a_{k'})$
- $\mathcal{C}(S)$ is the set of covers of S.
- M_k^i the set of non-stable matchings such that $\pi_i = a_k$

Second algorithm: centralized UCB

for $t = 1, \ldots, T$ **do**
  Run GS with players' rankings $\hat{\mu}_k^i + \sqrt{\frac{3\log(t)}{2T_k^i}}$
  Pull arm assigned by GS
end

Theorem (Regret bound)

$$\underline{R}_i(T) \lesssim \sum_{l:\underline{\Delta}_l^i > 0} \underline{\Delta}_l^i \min_{Q \in \mathcal{C}(M_l^i)} \sum_{(p_j, a_k, a_{k'}) \in Q} \frac{\log(T)}{(\mu_k^j - \mu_{k'}^j)^2}$$

where $\underline{\Delta}_l^i = \mu_{\underline{\pi}_i}^i - \mu_l^i$.

If a_i prefers p_i the most and p_i prefers a_i the most:

$$R_i(T) \lesssim \sum_l \frac{\log(T)}{\underline{\Delta}_l^i}.$$

Centralized UCB: pros and cons

Advantages:

- require no knowledge of problem parameters (e.g. Δ)
- scale with $\frac{1}{\Delta}$ in good cases
- incentive compatible (for unique stable matching at least)

Drawbacks:

- centralized GS algorithm
  $\rightarrow$ used at each round, so we cannot use decentralized GS instead
- pessimal sublinear regret only. Why?
  $\rightarrow$ see example later

Sketch of proof

- $\underline{R}_i(T) \leq \sum_{l:\underline{\Delta}_l^i > 0} \underline{\Delta}_l^i \left(\sum_{m \in M_l^i} T_m \right)$
- Define $L_{j,k,k'} = \sum_{m \in B_{j,k,k'}} T_m$ the number of times agent p_j pulls arm a_k when the triplet $(p_j, a_k, a_{k'})$ is blocking
- Hoeffding $\implies \mathbb{E}\left[L_{j,k,k'}\right] \lesssim \frac{\log(T)}{(\mu_k^j - \mu_{k'}^j)^2}$
- Conclude for every cover of M_l^i

Why pessimal regret?

Centralized UCB **does not** achieve sublinear optimal regret
Consider 3 players and 3 arms with

$$p_1 : a_1 \succ a_2 \succ a_3 \qquad a_1 : p_2 \succ p_3 \succ p_1$$
$$p_2 : a_2 \succ a_1 \succ a_3 \qquad a_2 : p_1 \succ p_2 \succ p_3$$
$$p_3 : a_3 \succ a_1 \succ a_2 \qquad a_3 : p_3 \succ p_1 \succ p_2$$

Optimal matching $\rightarrow (p_1, a_1), (p_2, a_2), (p_3, a_3)$

Why pessimal regret?

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$$p_2 : a_2 \succ a_1 \succ a_3 \qquad a_2 : p_1 \succ p_2 \succ p_3$$
$$p_3 : a_1 \succ a_3 \succ a_2 \qquad a_3 : p_3 \succ p_1 \succ p_2$$

Optimal matching $\rightarrow (p_1, a_1), (p_2, a_2), (p_3, a_3)$

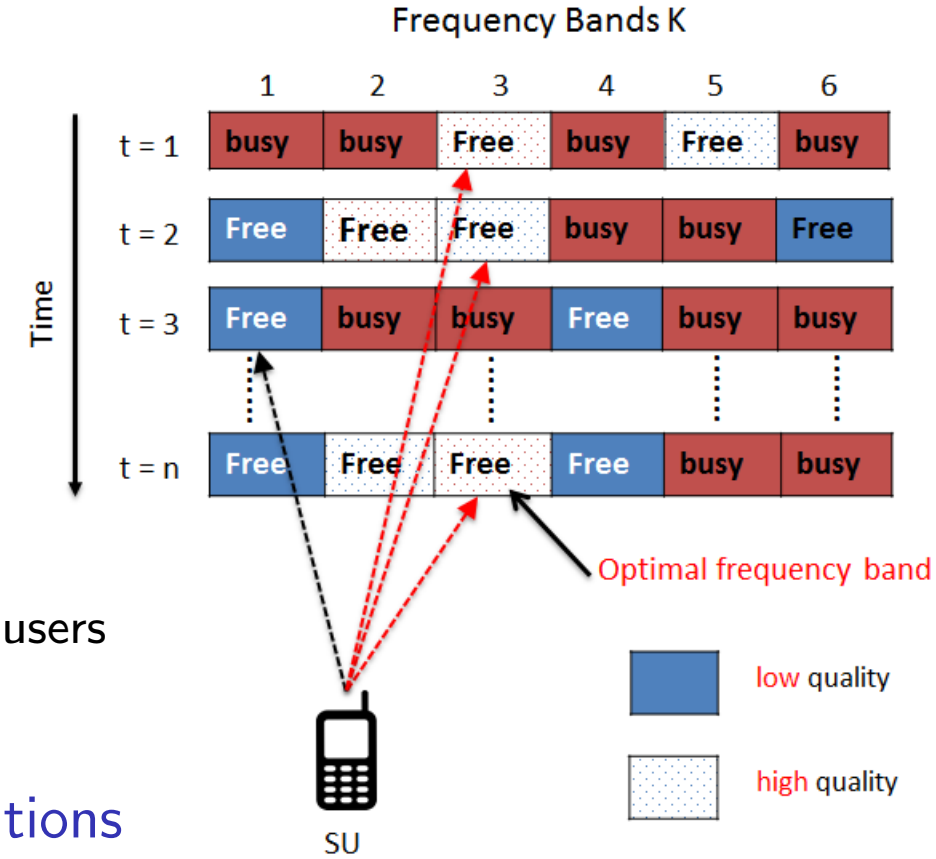
If p_3 overestimates a_1 and ranks $a_1 \succ a_3$
GS outputs $\rightarrow (p_1, a_2), (p_2, a_1), (p_3, a_3)$
Mistake never gets corrected

First conclusion

- nice framework combining learning and two sided markets
- simple algorithms with regret guarantees
- still many limitations
  - need for decentralized algorithm without knowledge of Δ
  - could it reach optimal regret ?
  - better dependencies in problem parameters

Link with Multiplayer bandits

Motivation: Cognitive Radio



- licensed bands: Opportunistic Spectrum Access

arm $\leftrightarrow$ availability of channel from primary users

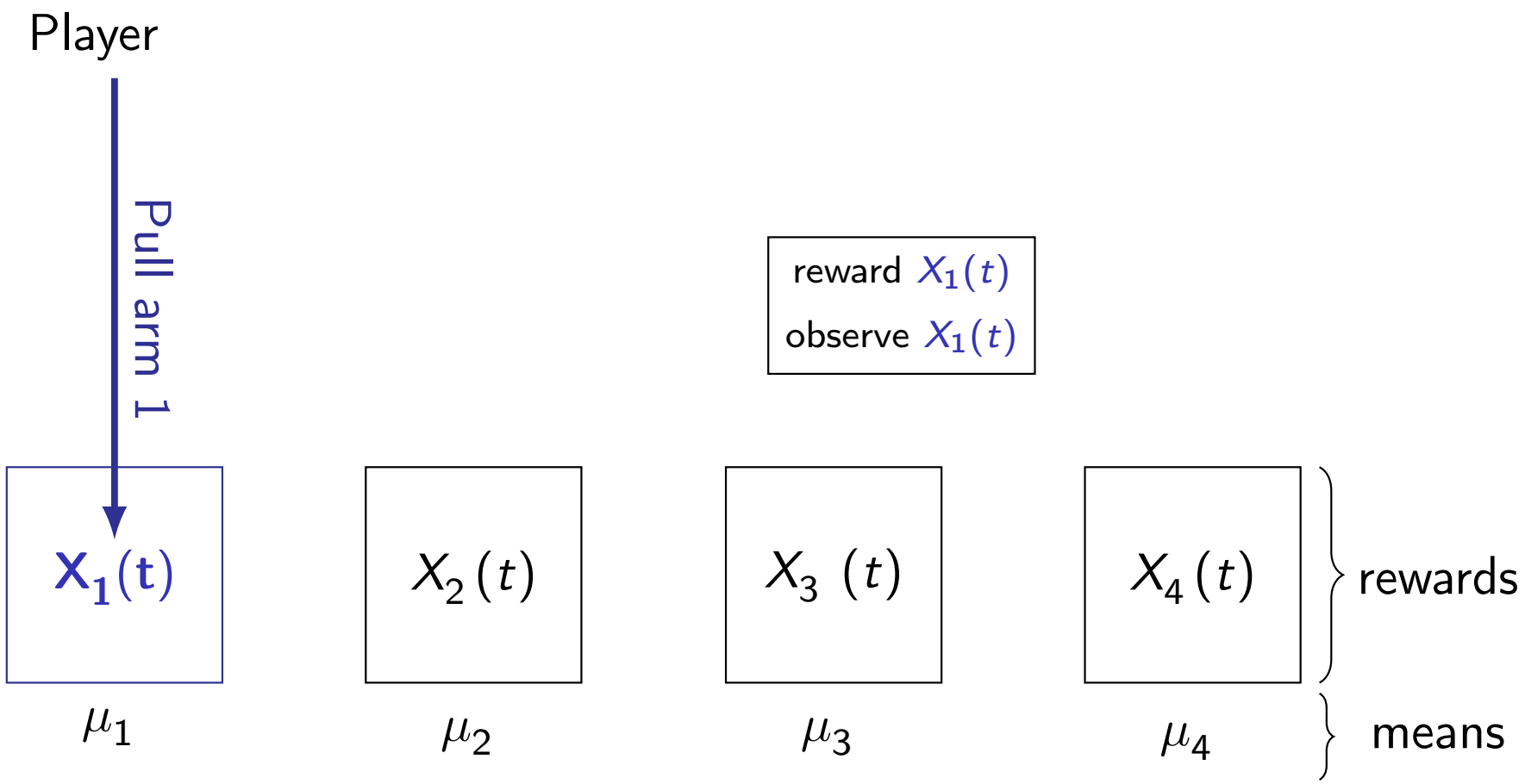
- un-licensed bands: IoT communications

arm $\leftrightarrow$ background traffic

$\rightarrow$ what about **multiple devices**?

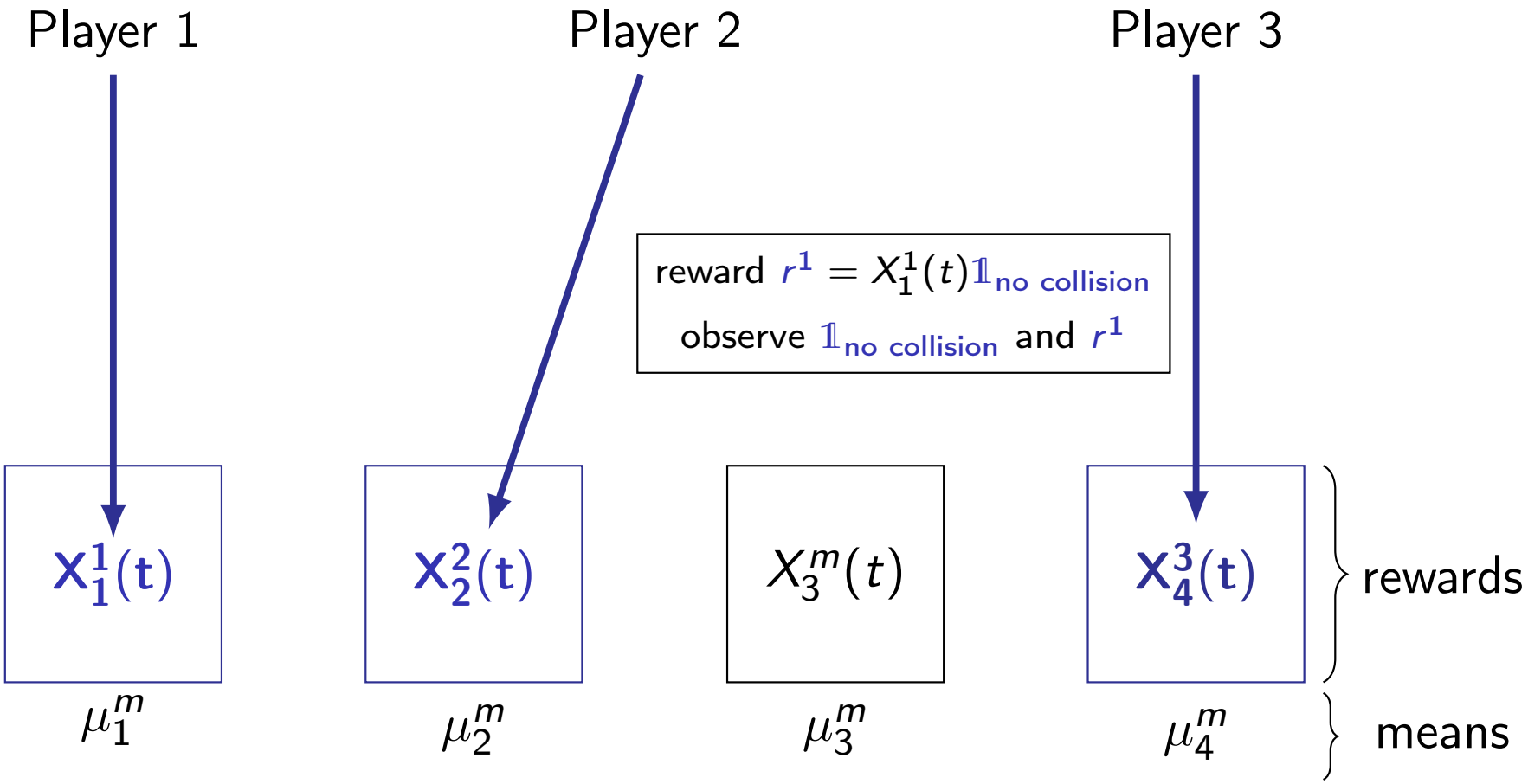
Stochastic bandits

K arms



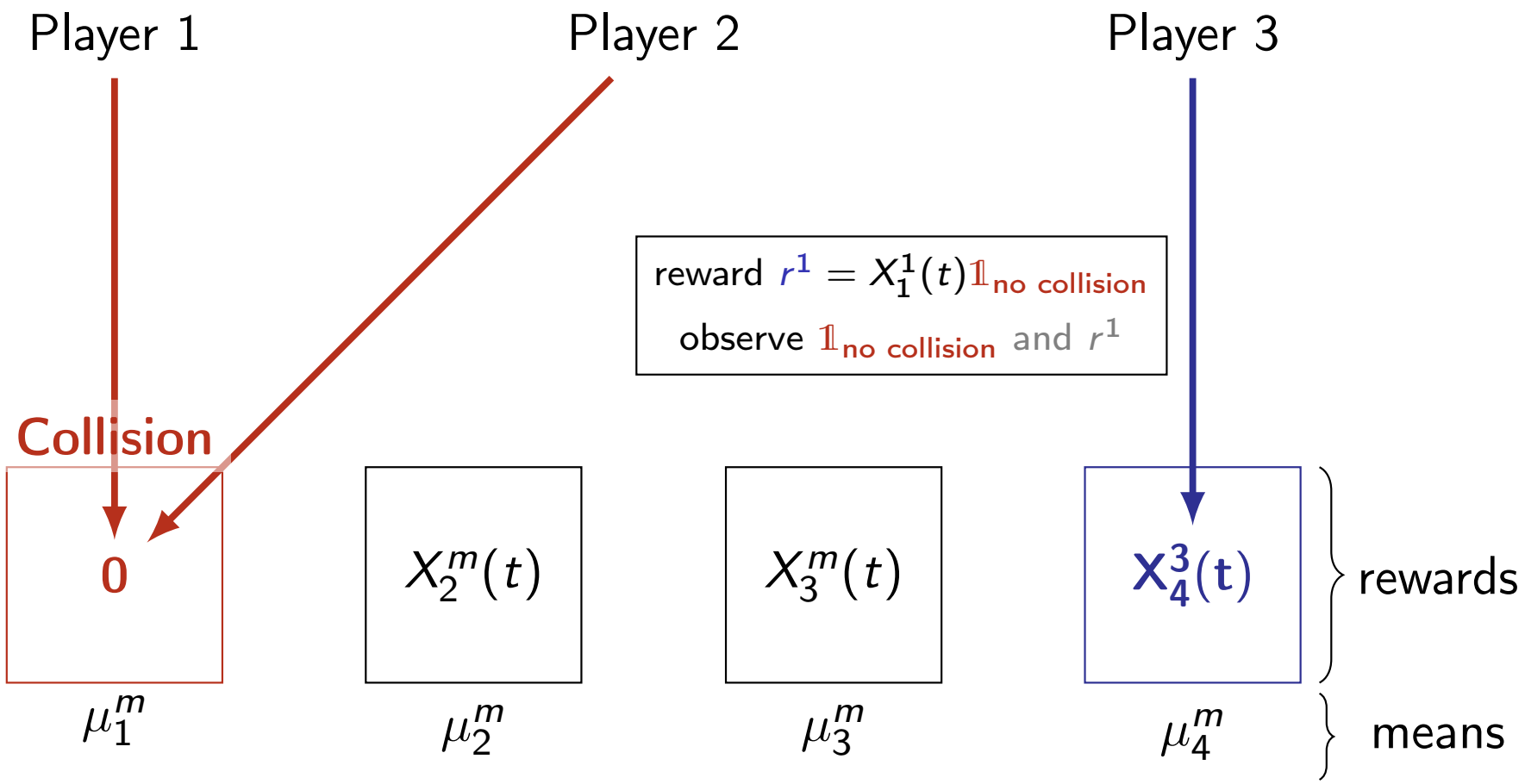
Stochastic bandits [**Multiplayer**]

K arms, M players



Stochastic bandits [**Multiplayer**]

K arms, M players



Interest for matching bandits

- motivated by cognitive radio networks
- growing literature on multiplayer bandits
  $\rightarrow$ survey to appear soon
- similarities, but different collision model
- incentive compatibility in recent work

Interest for matching bandits

Recent literature: suggested to use collision information $\mathbb{1}$ (no collision) as a bit sent between players [4]
$\rightarrow$ allow some level of communication

can it be used in matching markets?

[4] see e.g. Boursier E., Kaufmann E., Mehrabian A. and Perchet V. A practical algorithm for multiplayer bandits when arm means vary among players. *AISTATS 2020.*

Dominate or Delete: Decentralized Competing Bandits with Uniform Valuation

by Abishek Sankararaman, Soumya Basu & Karthik Abinav Sankararaman

Simplifications

Key Assumption: players have global ranks

$$p_1 \succ_k p_2 \succ_k \ldots \succ_k p_N \text{ for any } k.$$

order of players a priori known

Consequence

- unique stable matching
- computed with successive dictatorships

Collision information for player p_i:

$$\eta_i(t) = \begin{cases} 1 & \text{if } i = \min\{j \in [K] \mid \pi_j(t) = \pi_i(t)\} \\ 0 & \text{otherwise.} \end{cases}$$

Dominate or Delete: algo

Input: player rank i
$\mathcal{B}^i \leftarrow \emptyset$ `// blocked arms`
for $n = 0, \ldots, \infty$ **do**
  for 2^n *timesteps* **do** pull $\arg\max_{k \in [K] \setminus \mathcal{B}^i} \hat{\mu}_k^i + 2\sqrt{\frac{\log(t)}{N_k^i}}$
  `// UCB on non-blocked arms`
  Declare/detect most pulled arm for this epoch `// communication`
  $\mathcal{B}^i \leftarrow$ arms declared by players with rank $< i$ `// blocked arms`
end

Communication step done in time NK using collision information

Results

π_i^* arm of p_i in stable matching
$\mathcal{A}_i^* = \mathcal{K} \setminus \{a_{\pi_1^*}, \dots, a_{\pi_{i-1}^*}\}$ available arms for p_i
$\Delta_k^i = \mu_{\pi_i^*}^i - \mu_k^i$

$$R_i(T) \lesssim \overbrace{\sum_{l<i} \sum_{k \in \mathcal{A}_i^*} \frac{\log(T)}{(\Delta_k^l)^2}}^{\text{collisions with smaller ranks}} + \overbrace{\sum_{k \in \mathcal{A}_i^* \setminus \{\pi_i^*\}} \frac{\log(T)}{\Delta_k^i}}^{\text{exploration of } \mathcal{A}_i^*}$$

+ incentive compatible

comparable to centralized UCB in practice

Lower bound

$$R_i(T) \gtrsim \max\left\{ \frac{(j-1)\log(T)}{\Delta^2}, \frac{K\log(T)}{\Delta} \right\}$$

Bandit Learning in Decentralized Matching Markets

by Lydia T. Liu, Feng Ruan, Horia Mania & Michael I. Jordan

Conflict-Avoiding UCB

Global Observation: players observe who gets to pull a_k for all $a_k \in \mathcal{K}$
Assume the arms preferences $\succ_k$ are known to players

Input: player p_i, parameter λ
$k \sim \mathcal{U}(\mathcal{K})$
for $t = 1, \ldots, T$ **do**
  $D_i \sim \mathrm{Ber}(\lambda)$ **if** $D_i = 0$ **then**
    $\mathcal{B}^i \leftarrow \{a_k \mid p_j \succ_k p_i \text{ where } \pi_{p_j}(t-1) = a_k\}$ `// blocked arms`
    $k \leftarrow \arg\max_{a \in \mathcal{K} \setminus \mathcal{B}^i} \hat{\mu}_k^i + \sqrt{\frac{3\log(t)}{2N_k^i}}$ `// UCB`
  Pull k
end

Regret bounds

Case 1: globally ranked players
$\lambda = 0$, similar to Dominate or Delete

$$R_i(T) \lesssim i^2(K-i)\frac{\log(T)}{\Delta^2} + i^3 \sum_{k \in \mathcal{A}_i^* \setminus \{\pi_i^*} \Delta_k^i \frac{\log(T)}{\Delta^2}$$

Case 2: general case
$\lambda > 0, \varepsilon = (1-\lambda)\lambda^{N-1}$

$$\underline{R_i}(T) \lesssim \frac{N^5 K^2}{\varepsilon^{N^4}} \frac{\log^2(T)}{\Delta^2}$$

General case: how does this work?

Assume players correctly estimate their arm preferences
With proba ε, player p_i is only player trying to climb in preferences ($D_i = 0$)
$\rightarrow$ resolve a blocking pair (p_i, a_j)

Theorem (Abeledo and Rothblum, 1995)

Given any unstable matching π_0, there is a sequence of at most N^4 blocking pairs such that it yields a stable matching when resolved.

$\implies$ with proba ε^{N^4}, reach a stable matching in N^4 steps

Exponential term: a *perfect* sequence happens with small proba
better result possible? e.g. we are resolving blocking pairs in average

Conclusion

Good algorithms for specific case of global ranks

General case remains unsolved

- good centralized algorithm
- decentralized algorithm with central observation
  - exponential dependence in N
  - $\log^2(T)$ regret
  - incentive compatible?
- possibility of decentralized algo?
  $\rightarrow$ multiplayer bandits ideas might help, but not obvious to implement because asymmetric communication here